Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/42**

Paper 4 Alternative to Practical **May/June 2020**

MARK SCHEME

Maximum Mark: 60

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

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5 ‘List rule’ guidance

For questions that require ***n*** responses (e.g. State **two** reasons …):

- The response should be read as continuous prose, even when numbered answer spaces are provided
- Any response marked *ignore* in the mark scheme should not count towards ***n***
- Incorrect responses should not be awarded credit but will still count towards ***n***
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states ‘show your working’.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>A burette (1)
B measuring / graduated cylinder (1)
C pipette (1)</td><td>3</td></tr>
<tr><td>1(b)(i)</td><td>to provide H^+</td><td>1</td></tr>
<tr><td>1(b)(ii)</td><td>MnO_4^-</td><td>1</td></tr>
<tr><td>1(b)(iii)</td><td>
<table>
<tr><td>titration number</td><td>1</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>final burette reading / cm^3</td><td>17.2</td><td>34.1</td><td>17.1</td><td>16.9</td></tr>
<tr><td>initial burette reading / cm^3</td><td>0.0</td><td>17.2</td><td>0.7</td><td>0.2</td></tr>
<tr><td>volume of 0.600 mol / dm^3 hydrochloric acid</td><td>17.2</td><td>16.9</td><td>16.4</td><td>16.7</td></tr>
<tr><td>best titration results (✓)</td><td></td><td>✓</td><td></td><td>✓</td></tr>
</table>
final and initial readings correct in column 3 (1)
volume used correct in column 3 (1)
columns 1 and 4 correct (1)</td><td>3</td></tr>
<tr><td>1(b)(iv)</td><td>ticks in columns 2 and 4 and 16.8 (cm^3)</td><td>1</td></tr>
<tr><td>1(c)(i)</td><td>$0.0000845/8.45 \times 10^{-5}$ (mol)</td><td>1</td></tr>
<tr><td>1(c)(ii)</td><td>$0.0004225/4.2225 \times 10^{-4}$ (mol)</td><td>1</td></tr>
<tr><td>1(c)(iii)</td><td>0.0237 g (1)

0.237 g (1)</td><td>2</td></tr>
<tr><td>1(c)(iv)</td><td>12.9 (%)</td><td>1</td></tr>
</table>

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Question	Answer	Marks
2(a)(i)	green precipitate / ppt	1
2(a)(ii)	carbon dioxide (1) CO_3^{2-} / carbonate (1)	2
2(a)(ii)	(solution contains) SO_4^{2-} / sulfate ions	1
2(b)	dissolve the sample in distilled water **test for Ca^{2+}** add aqueous sodium hydroxide and warm (1) white ppt, insoluble in excess confirms Ca^{2+} (1) **test for NH_4^+** add aqueous sodium hydroxide and warm test gas with moist red litmus – turns blue to confirm NH_4^+ (1) **test for NO_3^-** add aqueous sodium hydroxide and warm in excess / until no more ammonia evolved (1) add aluminium and warm (1) test gas with moist red litmus – turns blue to confirm NO_3^- (1)	6

Question	Answer	Marks
3(a)	(fractional) distillation (1) ethanol and water have different boiling points (so ethanol boils off first) (1)	2
3(b)(i)	add water and stir / heat and filter (1) sand is residue / salt solution is filtrate **or** salt soluble / sand insoluble (1) heat filtrate to evaporate water (1)	3

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Question	Answer	Marks
3(b)(ii)	spot onto filter / chromatography paper (1) solvent / water in beaker and paper into solvent / water with spot above water level (1) R_f values different / travel different distances (1)	3

Question	Answer	Marks
4(a)	warm / heat gently	1
4(b)	keep adding CuO until in excess / black solid remains (1) to react with all the acid (1)	2
4(c)	filter (1) to remove excess CuO / solid (1)	2
4(d)	heat until solution is concentrated **or** saturated **or** until crystals start to form (1) leave solution to evaporate slowly (1) dry crystals on filter paper (1)	3
4(e)	**Any two from** (step 2) black solid remains (1) solution turns blue (1) (step 3) blue crystals (1)	2
4(f)	**Any one from** acid corrosive – suitable precautions heating – suitable precautions copper salts toxic – suitable precautions	1

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Question	Answer	Marks
5(a)	pink / orange / brown / red-brown / red-pink solid / layer on Zn **or** blue solution becomes paler blue / colourless	**1**
5(b)(i)	0.95 (g)	**1**
5(b)(ii)	the zinc was added at 4 minutes	**1**
5(b)(iii)	large surface area / increase rate of reaction	**1**
5(b)(iv)	glass is not good insulator / heat can escape (1) lag / insulate / put lid on beaker or use polystyrene cup instead (1)	**2**
5(b)(v)	points plotted before 4th min and no point at 4th min (1) points plotted after 4th min (1)	**2**
5(b)(vi)	straight line through first four points extended to cross 4th min	**1**
5(b)(vii)	straight line through last six points extended back to cross 4th min	**1**
5(b)(viii)	Δ7.2 (^{o}C)	**1**
5(b)(ix)	756(J)	**1**
5(b)(x)	0.0125 (mol)	**1**
5(b)(xi)	**5(b)(ix)** / 0.0125 (1) ÷ 1000 (1) negative sign (1)	**3**
5(c)	heat loss / reaction incomplete (as the zinc is coated)	**1**

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